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2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

(Math)

Group – A

(Objective Type Questions)

(Compulsory)

1. Choose the correct answer of the following : $1 \times 5 = 5$

(a) The symmetric difference of A = {1, 2, 3, 4} and B = null set is :

(i) {1, 2, 3} (ii) Null set

(iii) {1, 2, 3, 4} (iv) None of these

(b) The remainder when $2^{20} - 1$ is divided by 41 is :

(i) 0 (ii) 1

(iii) 2 (iv) None of these

(c) If $y = \log(\cos e^x)$ then $e^{-x}\frac{dy}{dx}$ is :

(i) $-\cos e^x$ (ii) $-\sin e^x$

(iii) $-\tan e^x$ (iv) None of these

(d) The number of distinct equivalence classes corresponding to the relation 'Congruence modulo 5' on the set of integers are :

(i) 1 (ii) 5

(iii) 10 (iv) None of these

(e) If $f : R \to R$ and $g : R \to R$ are defined as $f(x) = x + 1$ and $g(x) = \sin(x)$, then $gof(x)$ equals :

(i) $\sin(x + 1)$ (ii) $\sin(x) + 1$

(iii) $x + \sin(x)$ (iv) None of these

2. Fill in the blanks : $1 \times 5 = 5$

(a) The order of the element '3' in Z_7 is _____.

(b) If A is a finite set having 9 elements, then the number of elements in the power set of $A \times A$ is _____.

(c) The gcd(12379, 3051) is _____.

A→P

(d) A non-singular square matrix must have a determinant not equal to _____.

(e) If $y = x^{n+1}$ then y_n is equal to _____.

Group – B

(Short-answer Type Questions)

Answer any **four** questions of the following : $5 \times 4 = 20$

3. Show that composition of functions is not necessarily commutative.

4. Solve the following linear equations using matrix method :

$x + y + z = 1, x + 2y + z = 2, x + y + 2z = 0$

5. Show that $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

6. Define an equivalence relation on a non-empty set A. If R is a relation on $N \times N$ defined by (a, b) R (c, d) iff $a + d = b + c$, then prove that R is an equivalence relation.

7. In a group G prove that the identity element is unique.

8. Prove that every square matrix can be expressed as the sum of a symmetric and a skew symmetric matrix.

Group – C

(Long-answer Type Questions)

Answer any **four** questions of the following :

$10 \times 4 = 40$

9. If $u = \log \frac{x^2 + y^2}{xy}$, verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.

10. If $u = x \sin(y - x)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{u}{x}$.

11. $y = \sin(m \sin^{-1} x)$. Show that :

    (a) $(1 - x^2)y_2 - xy_1 + m^2 y = 0$

    (b) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (m^2 - n^2)y_n = 0$

12. Prove that : $\left\{\bigcup_{i=1}^{n} A_i\right\}' = \bigcap_{i=1}^{n} A_i'$

13. Test the consistency and solve the following system of simultaneous linear equations by matrix method : $x - 3y - 2z = 6$, $2x - 4y - 3z = 8$, $-3x + 6y + 8z = -5$.

14. Use Taylor's theorem to expand e^{x+h} in powers of h.

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